Multiresolution Analysis and the Scaling Function

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Key Concepts

1) Multiresolution analysis implies access to both the coarse or low resolution and fine scale or high resolution views of the signal.

2) A scaling function is used to define a space of signals $\mathcal{V}_0$: all those that can be expressed as weighted sums of shifted scaling functions.

3) The space at resolution j, $\mathcal{V}_j$, is defined using shifted versions of the scaling function scaled by 2^j.

4) As j increases, the resolution of space $\mathcal{V}_j$ increases because the scaling functions that define the space are increasingly compressed.

5) Lower resolution signals must lie in the higher resolution signal space.

    a) Lower resolution spaces are nest within higher resolution spaces: $\ldots \subset \mathcal{V}_{j-1} \subset \mathcal{V}_j \subset \mathcal{V}_{j+1} \subset \ldots$

    b) The scaling functions for level j must be a weighted sum of the scaling functions for level $j+1$. Hence the original scaling function satisfies the multi resolution property

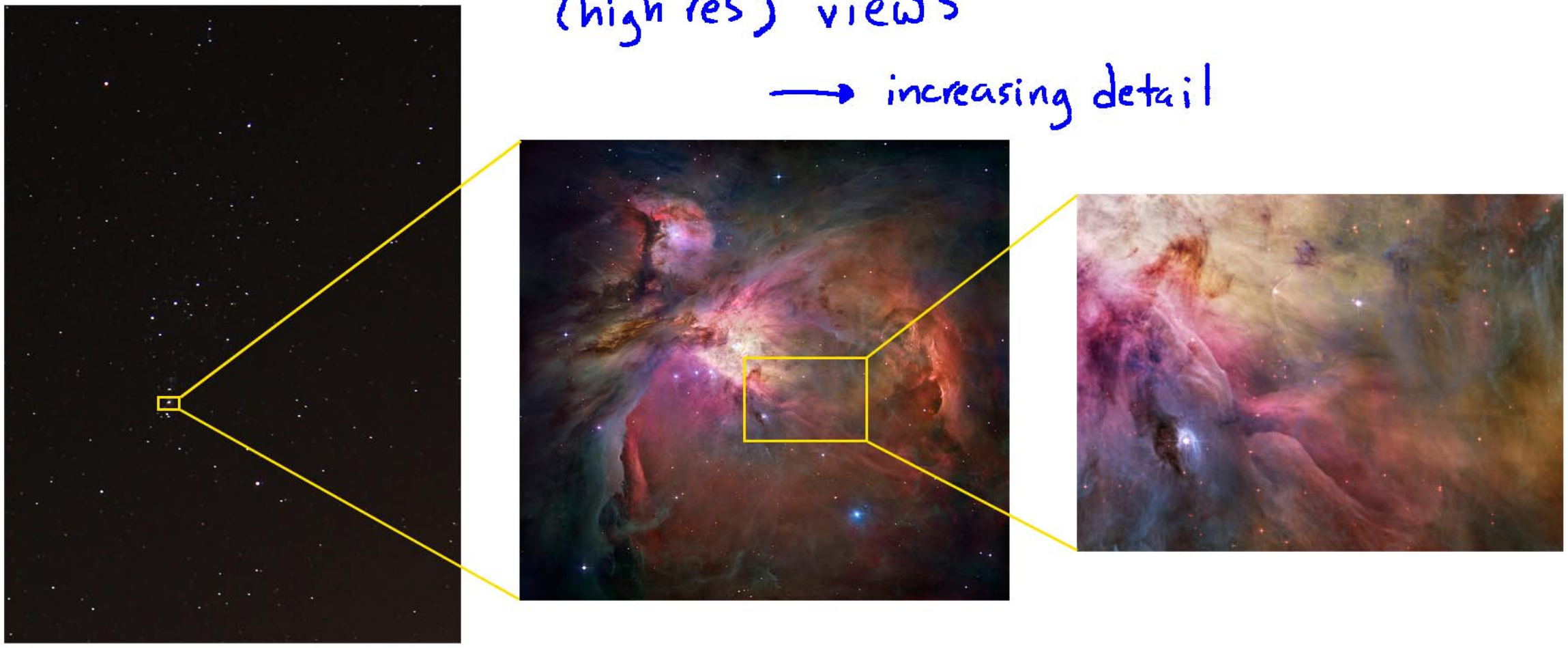
$$\phi(t) = \sum_n h_o[n] \sqrt{2} \phi(2t - n)$$

6) We also require the scaling function to be orthogonal to all other scaling functions.

Video Screenshots

Multiresolution - coarse (low res) and fine scale (high res) views

→ increasing detail

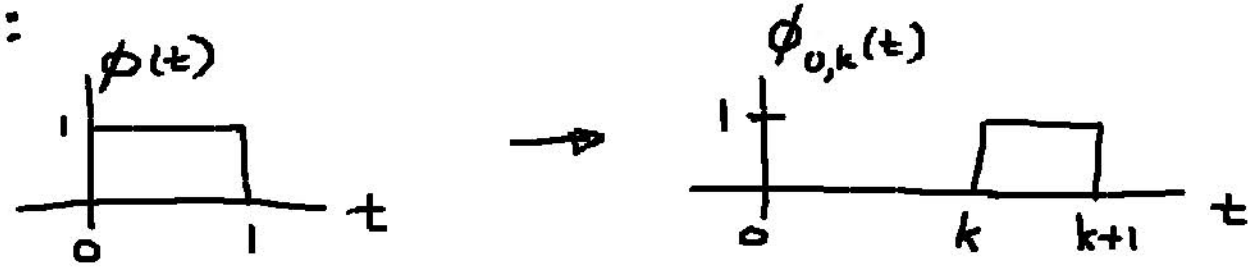


Wavelets analyze signals at multiple scales or resolutions

Scaling function $\phi(t)$

define $\phi_{0,k}(t) = \phi(t-k)$ for k integer

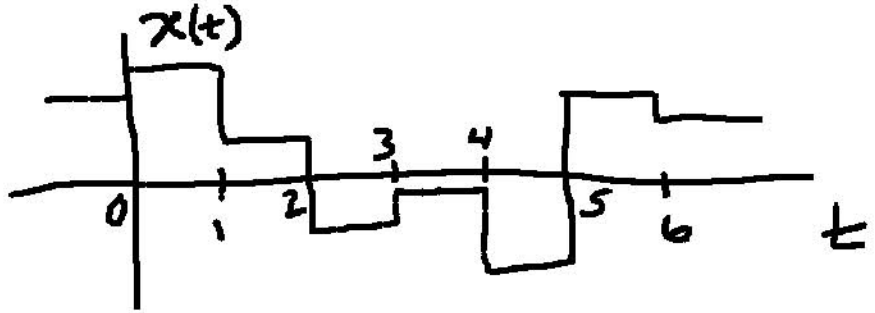
Example:



V_0 : space (set) of signals that can be obtained as weighted sums of $\phi_{0,k}(t)$

$$x(t) \in V_0 \Rightarrow x(t) = \sum_k a_k \phi_{0,k}(t)$$

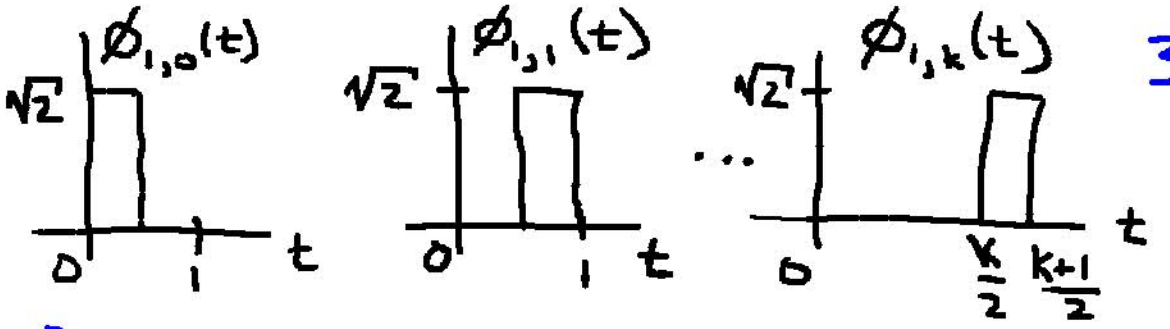
unit width stair steps



V_0 : base resolution "0"

$\phi_{1,k}(t) = 2^{1/2}\phi(2t-k)$

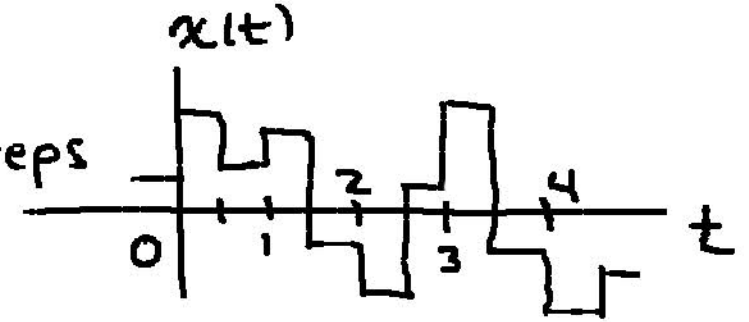
compress $\phi_{0,k}(t)$ by 2



V_1: $\{x(t) \mid x(t) = \sum_k a_k \phi_{1,k}(t)\}$

resolution "1"

Stairstep signals, 2^{-1} wide steps



$\phi_{j,k}(t) = 2^{j/2}\phi(2^j t-k)$

compress $\phi_{0,k}(t)$ by 2^j

V_j: $\{x(t) \mid x(t) = \sum_k a_k \phi_{j,k}(t)\}$

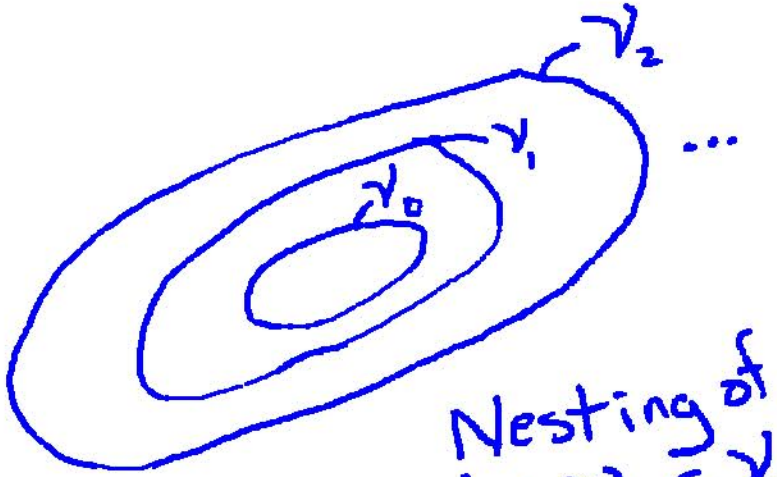
resolution "j"

Stairstep signals, steps 2^{-j} wide

As j ↑, $\phi_{j,k}(t)$ compresses and resolution ↑

If $x(t) \in V_j$, then $x(t) \in V_{j+1}$

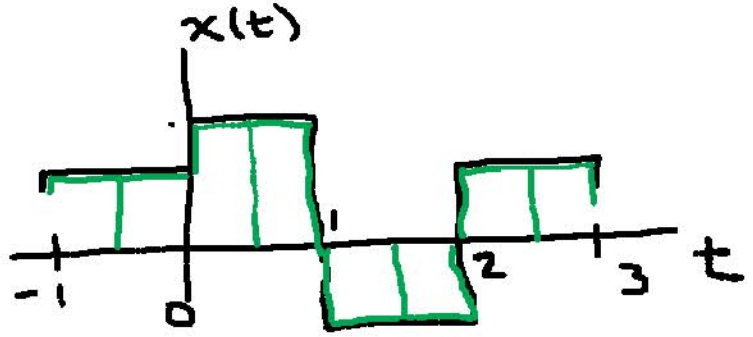
Lower resolution signals should be included in the higher resolution space



Nesting of spaces

$\ldots V_{-1} \subset V_0 \subset V_1 \subset V_2 \subset \cdots$

stairstep signals of width 1 are included in those of width 1/2



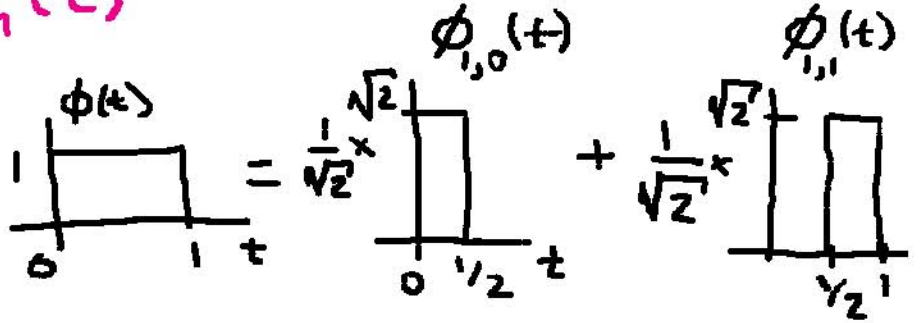
$x(t) \in V_0 \in V_1 \in \cdots$

The multiresolution property places an important constraint on the scaling function $\phi(t)$.

- If $V_j \subset V_{j+1}$, then $\phi_{j,k}(t) \in V_{j+1}$

$$\phi(t) = \sum_n h_0[n] \sqrt{2}\, \phi(2t-n) \qquad (j=0)$$

$$\phi_{0,0}(t) = \sum_n h_0[n]\, \phi_{1,n}(t)$$



Valid scaling functions must satisfy:

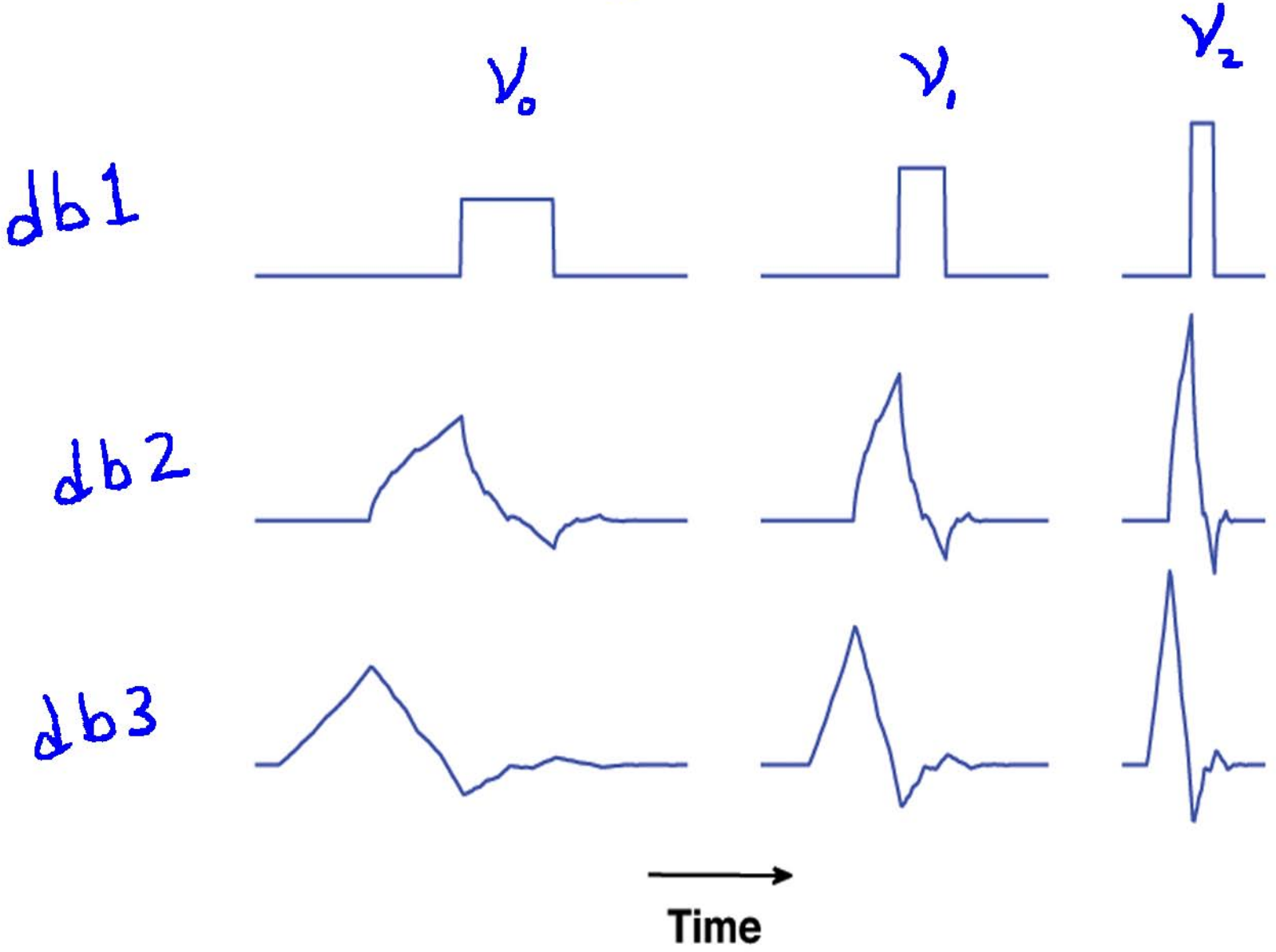
1) Orthogonality $\langle \phi_{j,k}(t), \phi_{\ell,m}(t) \rangle = 0$ $\ell \neq j$, $m \neq k$

$h_0[0] = \frac{1}{\sqrt{2}}$, $h_0[1] = \frac{1}{\sqrt{2}}$

2) Multiresolution property $\phi(t) = \sum_n h_0[n] \sqrt{2}\, \phi(2t-n)$

$h_0[n]$ important for defining and computing wavelet decomp

Example: Daubechies Scaling Functions



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